A120593 concise calculus certificate

VERIFIED

Bradley Klee, Mech.An.ika Sol (Open AI)

July 31, 2026

Definition and first terms

$5\,A(x) = 4 + 1\,x + A(x)^4$, with the origin-normalized solution.

Normalization/grammar: `A(x)=1+(1)*T(x); T=x+6*T^2+4*T^3+1*T^4.`

$a(0), \ldots, a(7) = 1, 1, 6, 76, 1201, 21252, 402892, 8001412.$

Kernel used throughout: $\rho(u) = u\,(1 - 6\,u^1 - 4\,u^2 - 1\,u^3)$, so $x = \rho(T(x))$.

Typogeometry

`1x1` `{1,1,0,0}x1` `{0,0,1,1}x1` `{1,1,1,0}x1`

literal 4-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: $\Delta_2 :\ 6,\ \Delta_3 :\ 4,\ \Delta_4 :\ 1.$

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \ldots, k_3) \in \mathbb{Z}^3_{\geq 0} : \sum_{j=1}^{3} j k_j = n - 1\}, \quad K = \sum_{j=1}^{3} k_j,$$

$$a(n) = \frac{1}{n} \sum_{\boldsymbol{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \ldots, k_3} 6^{k_1} 4^{k_2} 1^{k_3}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_\gamma \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^{3} P_r(n) a(n + r) = 0$, $n \geq 1$, with

$$P_0(n) = -256\,n^3 - 384\,n^2 - 48\,n + 40$$
$$P_1(n) = -3072\,n^3 - 9216\,n^2 - 8896\,n - 2752$$
$$P_2(n) = -12288\,n^3 - 55296\,n^2 - 79872\,n - 36864$$
$$P_3(n) = 491\,n^3 + 2946\,n^2 + 5401\,n + 2946$$

Differential equation

$$\left(40\,x^3\right) A(x) + \left(-688\,x^4 - 2752\,x^3\right) A'(x)$$
$$+ \left(-1152\,x^5 - 9216\,x^4 - 18432\,x^3\right) A^{(2)}(x) + \left(-256\,x^6 - 3072\,x^5 - 12288\,x^4 + 491\,x^3\right) A^{(3)}(x) = 0.$$

Matrices, telescoper certificate, and checks

$$U = \begin{pmatrix} \frac{11536}{491} & \frac{900}{491} & \frac{60}{491} & \frac{4}{491} \\ \frac{11040}{491} & \frac{736}{491} & \frac{180}{491} & \frac{12}{491} \\ \frac{3600}{491} & \frac{240}{491} & \frac{16}{491} & \frac{132}{491} \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad V = \begin{pmatrix} -1 & 0 & 0 & 0 \\ \frac{5644}{491} & \frac{409}{491} & \frac{60}{491} & \frac{4}{491} \\ \frac{3660}{491} & \frac{244}{491} & \frac{49}{491} & \frac{36}{491} \\ \frac{900}{491} & \frac{60}{491} & \frac{4}{491} & \frac{33}{491} \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Certificate.

$R(n, u) = N(n, u)/\rho(u)^2,\ \deg_u N = 9.$

Telescoping identity: $\sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u))$.

Matrix dimensions: 8×8; remainder 3×4, rank 3, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.

Canonical records: `data/tree_model.json`, `data/contour.json`, `data/matrices.json`, `data/recurrence.json`, `data/ode.json`. Embedded payload contains exact sources and SHA-256 matrix identifiers. Compare with the verbose certificate for A120590.